Deep Learning project 2 plan

Barbara Seweryn, Michał Wietecki

1. Project objective

The objective of this project is to compare and analyze the performance of different deep learning models in the speech classification problem. The team will experiment with different model parameters and investigate their influence on the model results. In the final stages.

2. Dataset

The data is sources from the Kaggle TensorFlow Speech Recognition Challenge. In contains the Speech Commands Dataset which can be used to build a model that understands and classifies simple spoken commands. This dataset contains 1-second speech recordings, each one classified as one of the following labels:
{“yes”, “no”, “up”, “down”, “left”, “right”, “on”, “off”, “stop”, “go”, “silence”, “unknown”}.

3. Models

In the project experimentation 3 models will be trained, evaluated and compared. The models we chose to investigate are:

- Baseline CNN,
- Transformer-based model (trained from scratch),
- Pretrained Transformer model.

For all of the models the influence of the learning rate parameter will be investigated. Additionally a set of parameters for investigation for each model will be chosen. At the moment the parameters we plan to investigate are:

- CNN - number of filters, dropout,
- Transformer – number of encoder layers, number of attention heads, dropout,
- Pretrained Transformer – fine-tuning strategy (freeze, partial, etc.).

If trouble with detecting the “silence” or “unknown” class will appear, a 2-stage approach will also be implemented, which will combine two models (selected from the best performing previously achieved models), the first one focusing on detecting silence, and the second one focusing on word classification.

4. Training environment

The preliminary experiments will be done on a subset of the data (2-4 classes, e.g. {“yes”, “no”, “up”, “down”} subset) and after investigating the influence of different model parameters and finding the optimal ones, the final models will be trained on the entire data sample. This adjustment is made purely for computational time reasons, since the team plans to perform the calculations using the Kaggle GPU, which has a 30 hours per week limit. All models will be trained using the same optimizer, batch size, loss function etc. to ensure a sensible comparison environment. To ensure experiment reproducibility, a random seed will be fixed.

5. Experimentation outline

Experimentation Stage	Subject	Details
1	Baseline training	The 3 base models will be trained and evaluated.
2	Parameter investigation	The proposed sets of parameters will be investigated. For each of the parameters a set of 2-3 values will be used for training and the model performance will be evaluated for each of them.
3	Choice of the best performing models	Based on the results from previous stage the parameter values with the best influence will be chosen.
4	Model training on the entire sample.	The models with the chosen parameters will be trained on the entire dataset and their performance will be evaluated and compared.
5	2-stage model	To improve robustness on the “silence” and “unknown” classes, a 2-stage approach will also be implemented, which will combine two models, the first one focusing on detecting silence, and the second one focusing on word classification.

6. Evaluation metrics

The models will be evaluated primarily using typical classification metrics, such as accuracy and F1-score. Training and validation loss will also be monitored over the epochs to analyze the learning dynamics, convergence speed and overfitting. Confusion matrices will be used to visualize and investigate which classes are commonly confused by the models, especially: confusion between phonetically similar words, performance on the “unknown” and “silence” classes.